Supplementary Material

Bootstrapping Video Interaction Generation with Synthetic State Transitions

A SVM Evaluator

	Precision	Recall	F1-score	Support
Bad	0.82	0.78	0.80	530
Good	0.61	0.65	0.63	268
Macro Avg	0.71	0.72	0.71	798
Weighted Avg	0.75	0.74	0.74	798
Accuracy		0.74		
ROC AUC		0.77		

Table 6: **Overall Our SVM Evaluator Performance.**

B Taxonomy Details

This section provides a more detailed description of the structure and content of the interaction taxonomy introduced in Section 3.1. This taxonomy is designed to enable the compositional generation of a wide and diverse range of interaction prompts, spanning from real-world interactions to creative scenarios.

The first core pillar of the taxonomy, ‘Actors/Objects’, defines the subjects and objects of interaction. It is broadly divided into ‘Animate’ and ‘Inanimate’ categories. The ‘Animate’ category includes hundreds of species of animals (including mammals, birds, insects, and dragons) and plants, while the ’Inanimate’ category hierarchically organizes a wide range of objects, from everyday items like furniture, vehicles, tools, and food, to fantasy items.

The second core pillar, ‘Interactions/Actions’, defines the possible events. Going beyond a simple list of verbs, we have semantically grouped actions according to their physical outcomes. For example, within the ’Physical Manipulation’ category, there are subgroups like ‘Deformation’ and ‘Separation/Fracture,’ which provide a structured basis for generating nuanced and specific state changes. We define approximately 500 detailed actions, including various interaction types such as spatial interactions, state changes, and creative/constructive actions.

Our taxonomy is designed around the core principles of comprehensiveness and composability. This enables our prompt generator to effectively explore a vast number of realistic and creative interaction scenarios by combining diverse elements within this structured space.

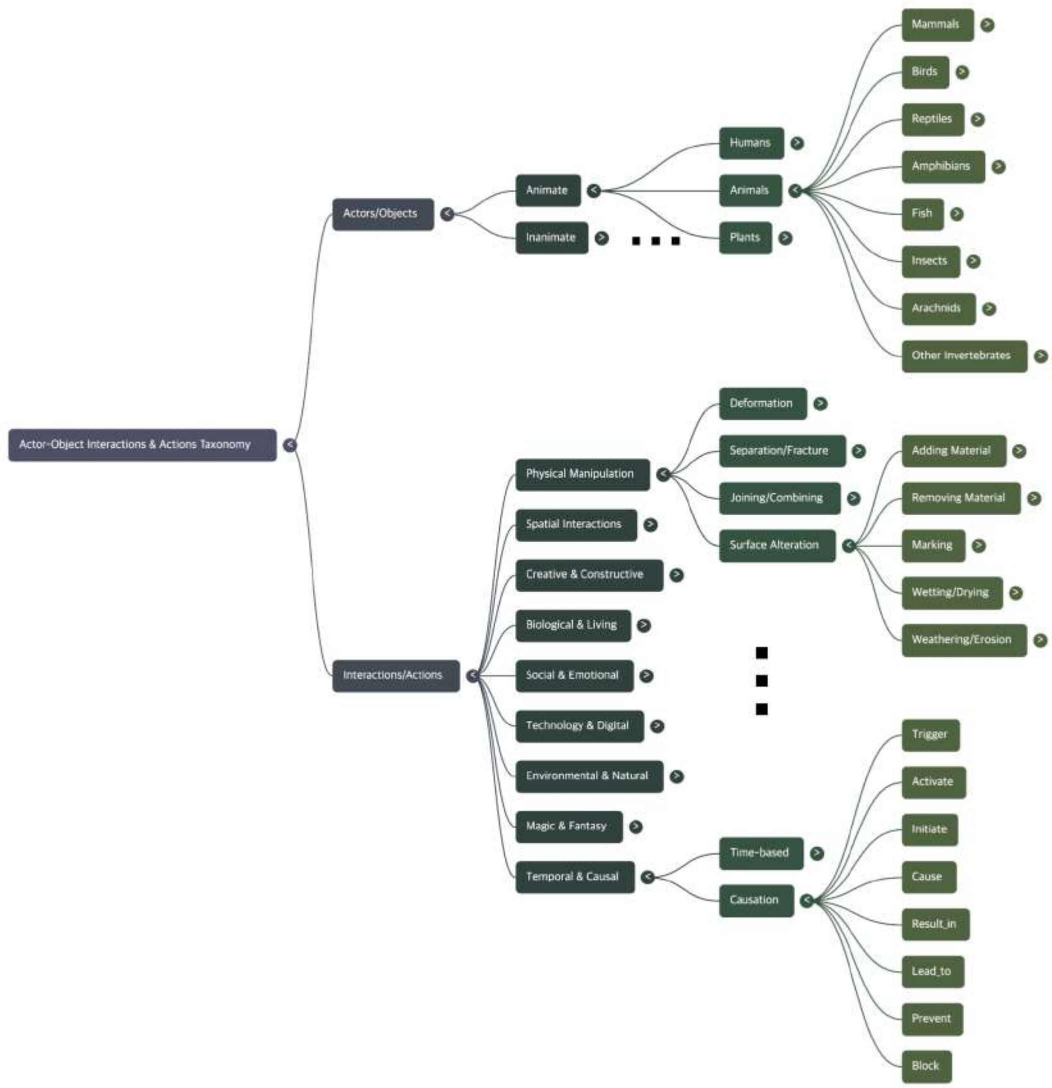
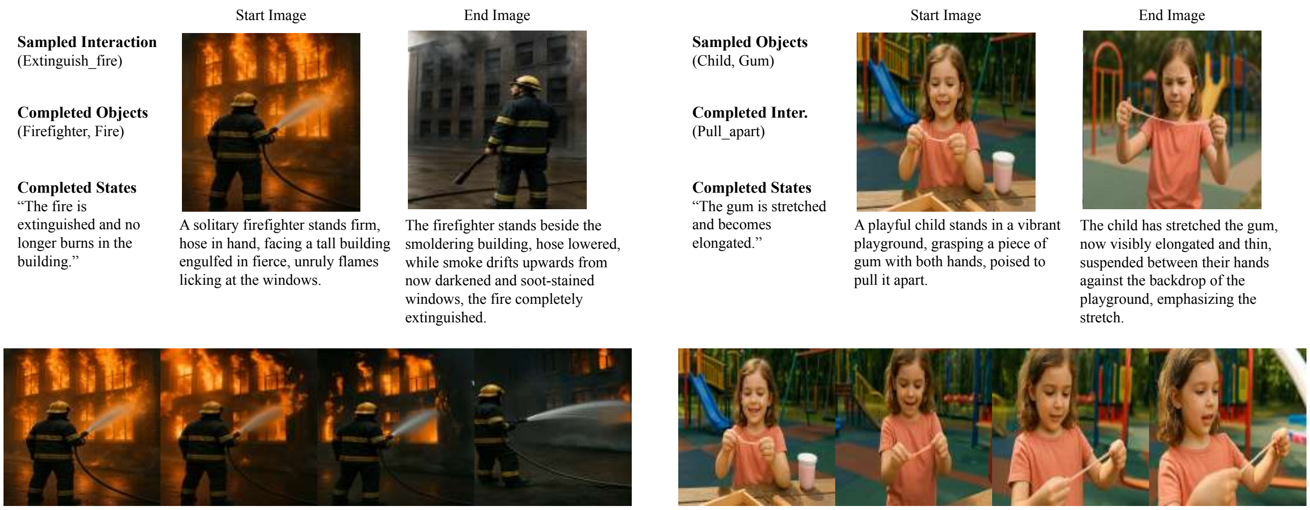


Figure 5: **The hierarchical structure of our Actor-Object and Interactions-Actions Taxonomy**, forming the basis for generating diverse interaction scenarios.



(a) Watch as a determined firefighter battles fierce flames with a powerful hose, transforming a burning building into a smoldering, smoke-filled structure.

(b) A child in a vibrant playground pulls apart a piece of gum, stretching it into a long, thin strand.

Figure 6: **An example of our data generation pipeline.** From a sampled interaction (Extinguish_fire) and objects (Child, Gum), our method completes the tuple, generates detailed prompts for the ‘start’ and ‘end’ states, synthesizes the corresponding images, and finally generates the video representing the state transition.

(a) Watch as a determined firefighter battles fierce flames with a powerful hose, transforming a burning building into a smoldering, smoke-filled structure.

(b) A construction worker swings a bat against a wooden bench, causing it to break apart into scattered pieces on the gravel.

Figure 7: **Failure cases** produced by naive sampling methods. These examples illustrate the typical artifacts that our State-Guided Sampling (SGS) is designed to resolve. (a) The I2V model fails to depict the state change, resulting in a static and visually inconsistent video. (b) The FLF model creates an abrupt and unnatural transition with noticeable visual artifacts.

C Generation and Evaluation Examples

C.1 Generated Examples

C.2 VLM evaluation examples

Gemini Evaluation on Fig. 6 (a).

1. Presence and Clarity of Interaction
Score: 5/5
Reason: The prompt describes a firefighter using a hose to battle flames. The video clearly and unambiguously depicts this entire process. The stream of water from the hose is visible, and its contact with the burning building is the central focus of the video, perfectly matching the prompt’s intent.
2. Physical Plausibility of Video
Score: 4/5
Reason: The video is mostly physically plausible. The way the water sprays from the hose, the generation of smoke/steam as it hits the fire, and the general dimming of the scene as the flames are extinguished are all relatively natural. The fire itself looks slightly artificial, but its reaction to the water is believable.
3. Interaction State Change Causality
Score: 5/5
Reason: The prompt explicitly calls for a transformation from a “burning building” to a “smoldering, smoke-filled structure.” The video perfectly illustrates this cause-and-effect relationship. It is unequivocally clear that the firefighter spraying water is the direct cause of the flames diminishing and being replaced by thick smoke.
4. Temporal Continuity and Absence of Unnatural Jumps
Score: 5/5
Reason: The video is exceptionally smooth. The movement of the firefighter, the flow of water, and the transition from fire to smoke all occur seamlessly with no noticeable jumps, flickers, or unnatural artifacts. The temporal flow is completely natural.
Total Score: 19/20

Gemini Evaluation on Fig. 6 (b).

Video Evaluation
Prompt: A child in a vibrant playground pulls apart a piece of gum, stretching it into a long, thin strand.
1. Presence and Clarity of Interaction:
Score: 5/5
Reason: The video perfectly depicts the interaction described in the prompt. A child is clearly shown holding a pink, stretchy substance (consistent with gum or slime) and pulling it apart with both hands. The entire process of stretching is the central action and is shown unambiguously.
2. Physical Plausibility of Video:
Score: 5/5
Reason: The physics of the interaction are highly plausible. The way the substance stretches, thins in the middle, and resists the pull is very consistent with the properties of a material like gum or slime. The child's movements and the static environment are all natural.
3. Interaction State Change Causality:
Score: 5/5
Reason: The cause-and-effect relationship is perfectly clear. The action of the child pulling her hands apart (the cause) directly and visibly results in the substance changing its state from a small lump to a long, thin strand (the effect). The link is immediate and undeniable.
4. Temporal Continuity and Absence of Unnatural Jumps:
Score: 5/5
Reason: The video is completely smooth and continuous. There are no noticeable jumps, flickers, or unnatural movements. The action flows naturally from beginning to end.
Total Score: 20/20
Overall Comments: This is an excellent video that perfectly matches the prompt's intent. The interaction is clear, physically believable, and demonstrates a perfect cause-and-effect relationship. The technical quality is also very high, with smooth and natural motion.

Gemini Evaluation on failure case Fig. 7 (a).

Evaluation Results:
1. Presence and Clarity of Interaction: 4/5
Reasoning: The video clearly depicts the interaction specified in the prompt: a firefighter spraying a hose at a burning building. The firefighter, hose, water stream, and fire are all present and their relationship is clear. The mechanism of interaction (water hitting the fire) is mostly clear, though the exact point of impact is somewhat obscured by the flames.
2. Physical Plausibility of Video: 2/5
Reasoning: The video has several physically implausible elements. The firefighter is completely static, showing no reaction or strain from holding a high-pressure hose. The water stream itself is also static, lacking the arc and spray pattern expected from real-world physics. The fire appears to be a simple looping animation and does not react dynamically to the water being sprayed on it.
3. Interaction State Change Causality: 1/5
Reasoning: This is the video's biggest failure. The prompt explicitly asks to see the firefighter's actions ẗransforming a burning building into a smoldering, smoke-filled structure.Ḧowever, the water being sprayed on the fire has absolutely no discernible effect. The flames do not diminish, change color, or produce more smoke. The state of the building does not change at all as a result of the interaction. The cause (spraying water) does not lead to any effect.
4. Temporal Continuity and Absence of Unnatural Jumps: 5/5
Reasoning: The video is perfectly smooth. It functions like a cinemagraph where certain elements (the fire) are animated in a loop while others remain static. There are no jumps, glitches, or unnatural warping, resulting in a very stable temporal flow.
Total Score: 12/20
Overall Comments: The video successfully establishes the scene and the intended interaction from the prompt. However, it completely fails to show the consequence of that interaction. It's more of a "living image" than a video depicting a cause-and-effect event. While temporally smooth, the lack of physical reaction and state change makes the core interaction feel pointless and unrealistic.

Gemini Evaluation on failure case Fig. 7 (b).

Video Evaluation
1. Presence and Clarity of Interaction: 2/5
Reasoning: The prompt specifies the worker "swings a bat against a wooden bench." The video shows the worker preparing to swing (0:00) and the immediate aftermath of the bench breaking (0:01), but the critical moment of the bat making contact with the bench is completely omitted. The interaction is implied by the result, but the mechanism of the interaction is not depicted, making the prompt's intent poorly reflected.
2. Physical Plausibility of Video: 1/5
Reasoning: The video has several major physical implausibilities. The bench doesn't break as if hit by a bat; it explodes outwards from its center as if a charge was detonated within it. Most significantly, a second, identical baseball bat appears on the ground from 0:02 onwards, while the worker is still holding the original bat. This is physically impossible.
3. Interaction State Change Causality: 2/5
Reasoning: While the bench does break (a state change), the causal link is very weak. Because the impact is not shown, the bench's destruction feels more like a sudden, spontaneous event that happens to coincide with the worker's action rather than being a direct result of it. The "cause" (the hit) is missing, making the "effect" (the break) feel disconnected.
4. Temporal Continuity and Absence of Unnatural Jumps: 1/5
Reasoning: The video suffers from severe temporal discontinuity. There is a major jump cut between 0:00 and 0:01, skipping the entire swing and impact action. Furthermore, an object (a second bat) teleports into the scene at 0:02, which is a drastic and unnatural jump in the video's state.
Total Score: 6/20

In designing our VLM-based evaluation pipeline, we initially experimented with using a structured JSON format to query the model and receive its scores. However, we empirically found that forcing the model to adhere to a rigid JSON schema significantly degraded its evaluation performance compared to using an unconstrained natural language prompt. Consequently, we adopted a more effective two-step process: we first prompt the model using natural language to elicit a detailed, free-form text evaluation, and then parse this natural language output to extract the final structured scores.

D Temporal Artifact Detector

Correlation	Accuracy	Precision	Recall	F1-score
0.637	0.865	0.572	0.820	0.674

Table 7: **Performance of the temporal artifact detector.** The Pearson Correlation is calculated against the raw (1-5) human-rated Continuity scores. The binary classification metrics are based on a threshold where a human *Continuity* score ≤ 2.0 defines the positive class.

To validate the performance of our proposed temporal artifact detector, we used the 1,146 human-annotated videos as ground truth. The detector's continuous prediction score showed a high Pearson correlation of 0.64 with the 1-5 human-rated Continuity scores.

Furthermore, we measured its binary classification performance on detecting videos with severe artifacts. We define the positive class ('artifact present') as videos with a human-rated Continuity score of 2.0 or lower. The classification performance is shown in Table 7. As seen in the table, our detector achieves a high Recall of 0.82, successfully identifying the majority of videos that contain actual artifacts. While its Precision of 0.57 indicates the presence of some false positives, the overall F1-score of 0.67 demonstrates reliable performance. The high Recall aligns with our primary goal for dataset filtering, where correctly identifying poor-quality samples is paramount. This result validates that the detector is effective enough to serve as an auxiliary feature in our evaluation framework.

E Additional Qualitative Results

Figure 8 provides a clear visual demonstration of the 'ghosting effect,' a critical artifact that arises from naive score mixing approaches, and illustrates how our State-Guided Sampling (SGS) method resolves it. The examples labeled 'Constant Interpolation' show the result of using a fixed, linear weight to combine the FLF and I2V models. This method's rigid adherence to the target end-frame forces an unnatural, translucent overlay of the start and end states, which is particularly prominent in the final frames of the sequence.

As shown in the right column, SGS effectively mitigates this artifact. By exponentially decaying the FLF model's influence towards the end of the sequence, SGS allows the I2V model's strength in maintaining local coherence to dominate. This results

Figure 8: **Qualitative examples of the 'ghosting effect'.** Naive Constant Interpolation (left column) results in an unnatural, translucent overlay in the final frames. In contrast, our proposed SGS (right column) resolves this artifact by dynamically adjusting model influence, producing a clear and temporally coherent final state.

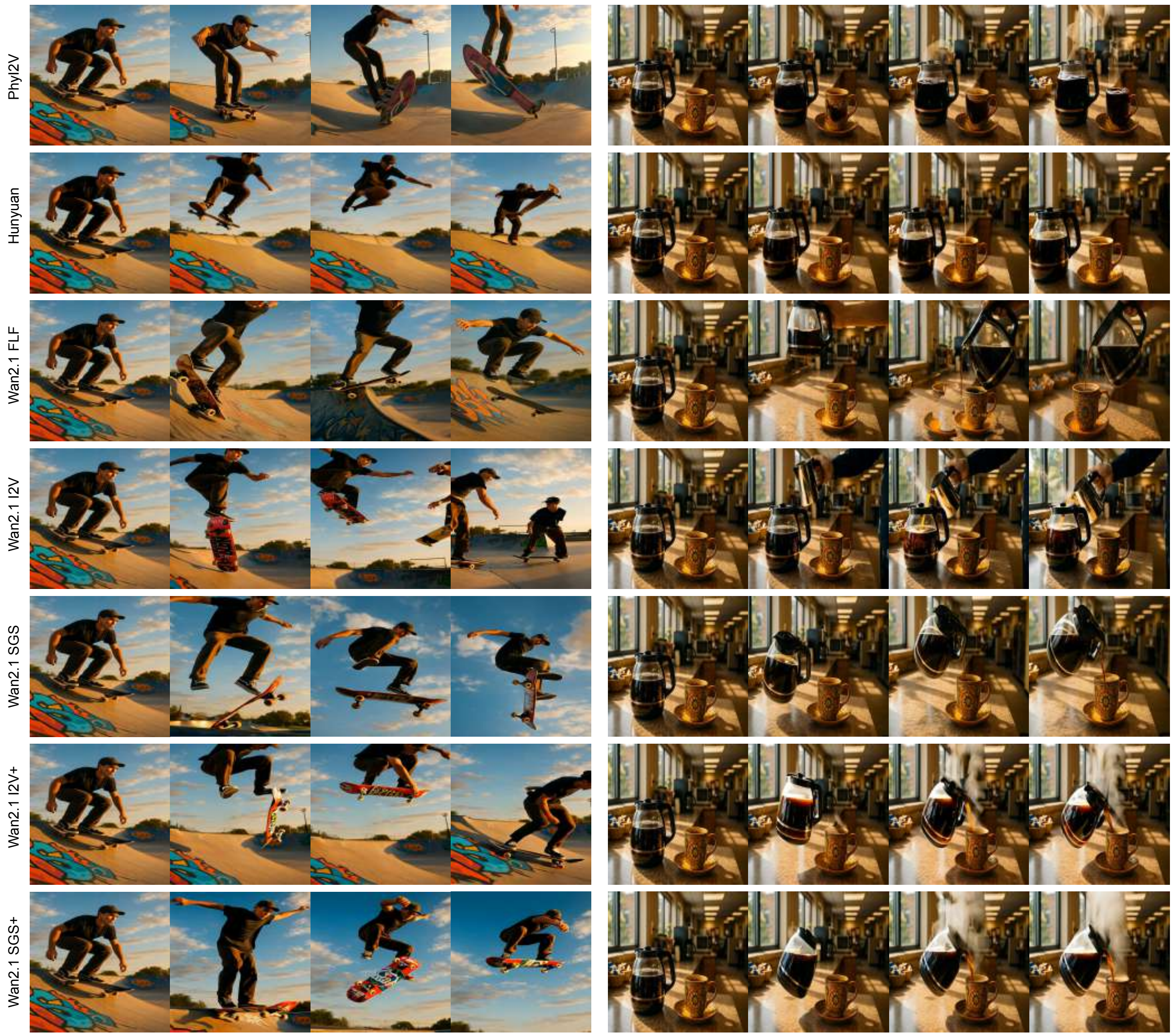
in a physically plausible and temporally consistent final state that naturally evolves from the preceding motion, highlighting the necessity of our dynamic weighting scheme.

In addition to this analysis, we provide further qualitative results, including direct comparisons with baseline models and additional examples contrasting the FLF, I2V, and SGS sampling methods.

(a) A child presses their finger firmly into the yellow modeling clay on the table, creating a clear indentation in its surface.

(b) The nightingale hops onto the coffee table and pecks at the cherry, puncturing its skin and partially eating the fruit, leaving small pieces scattered around.

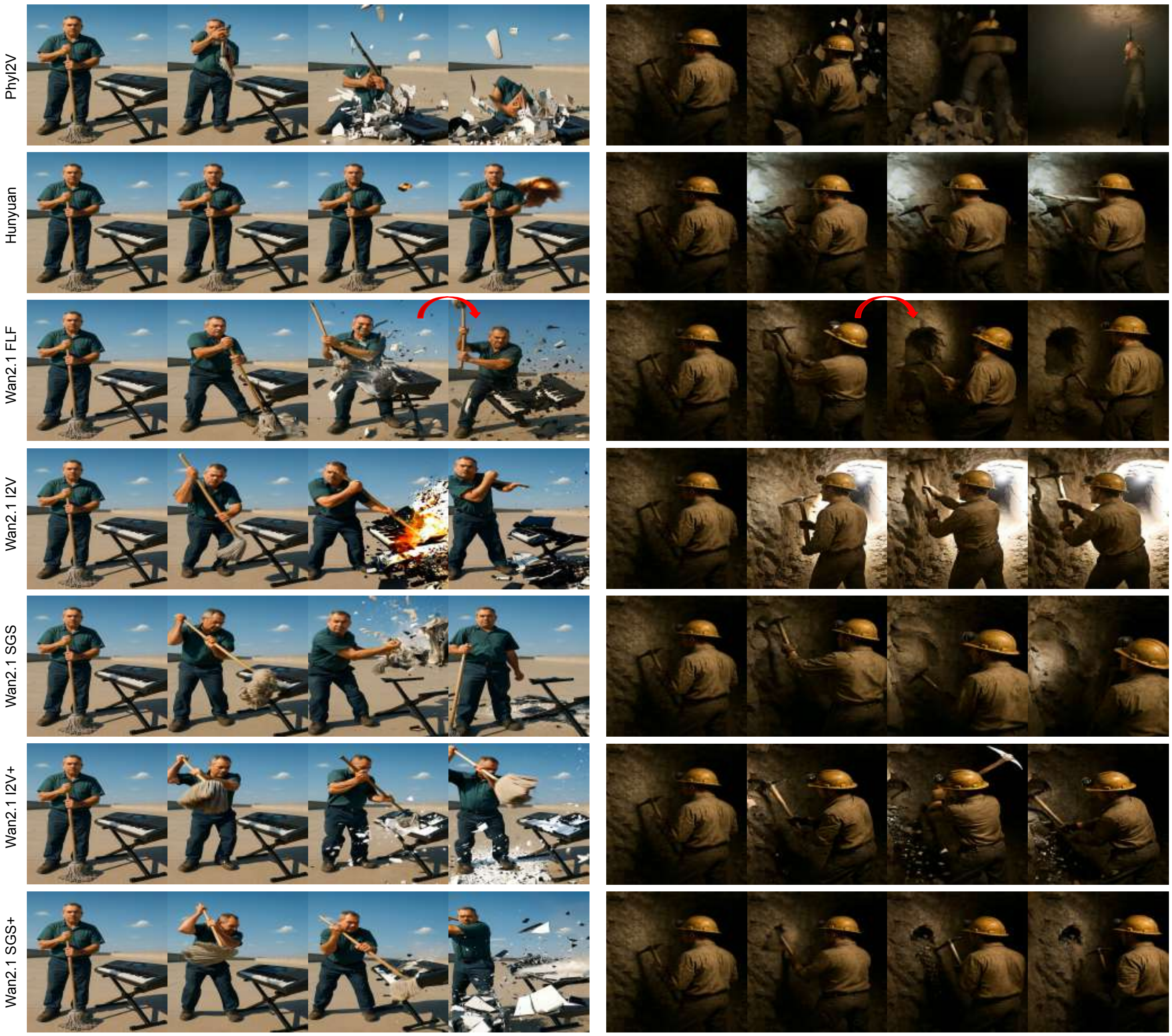
Figure 9: **Qualitative Baseline Comparison.**



(a) With a burst of energy, the skateboarder launches off the ramp, soaring upward as the skateboard spins beneath them in mid-air, both suspended in a dramatic, sunlit leap.

(b) The coffee pot tilts in midair and pours a swirling stream of steaming coffee into the vibrant mug, magically filling it to the brim in the sunlit office break room.

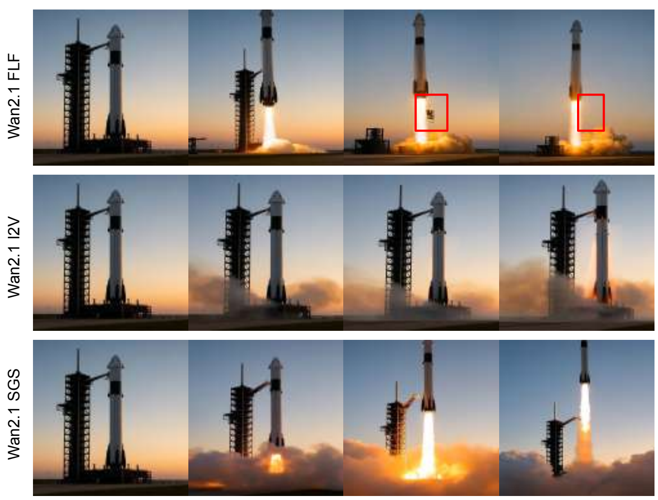
Figure 10: **Qualitative Baseline Comparison.**



(a) With a fierce swing of his mop, the janitor smashes the electronic keyboard, sending its keys and panels flying apart in a dramatic explosion of fragments.

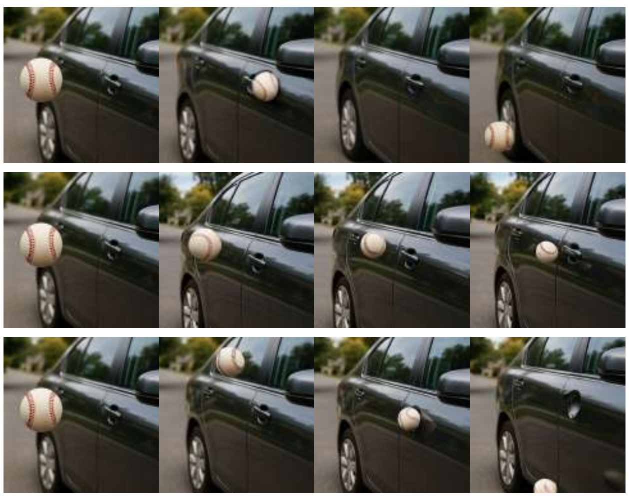
(b) A miner uses a pickaxe to break through the rock wall in an underground tunnel, creating a cavity and scattering debris on the ground.

Figure 11: **Qualitative Baseline Comparison.**

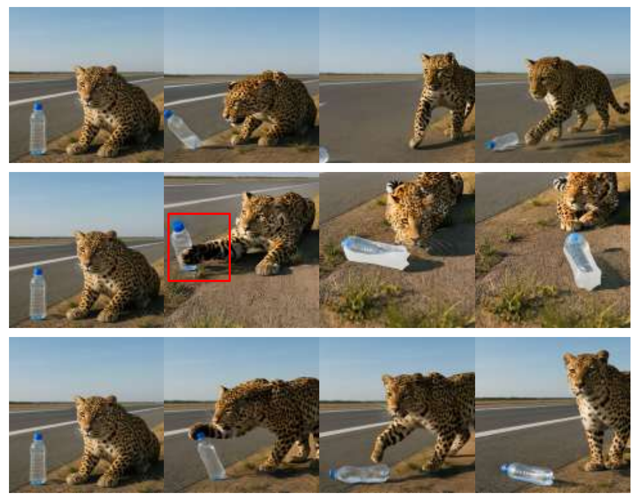


(a) Watch as the rocket ignites and propels the spacecraft upwards into the atmosphere, leaving the launch pad behind.

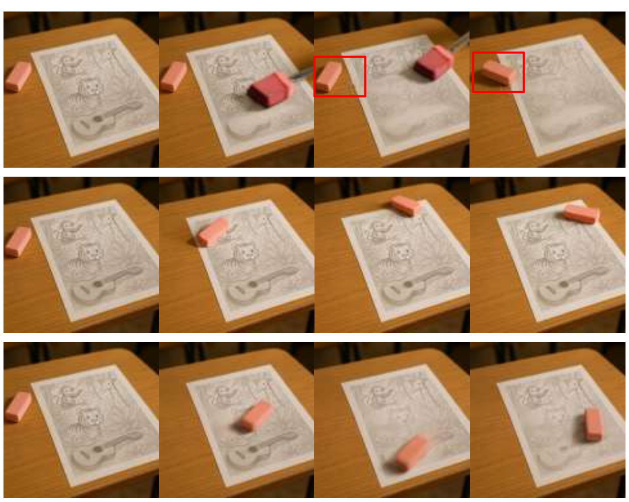
(b) A curious child in a kitchen eagerly takes bites out of a large red apple, reducing it to a half-eaten state with visible teeth marks.



(c) A baseball hurtles through the air, striking the car door and leaving a noticeable dent in its center.



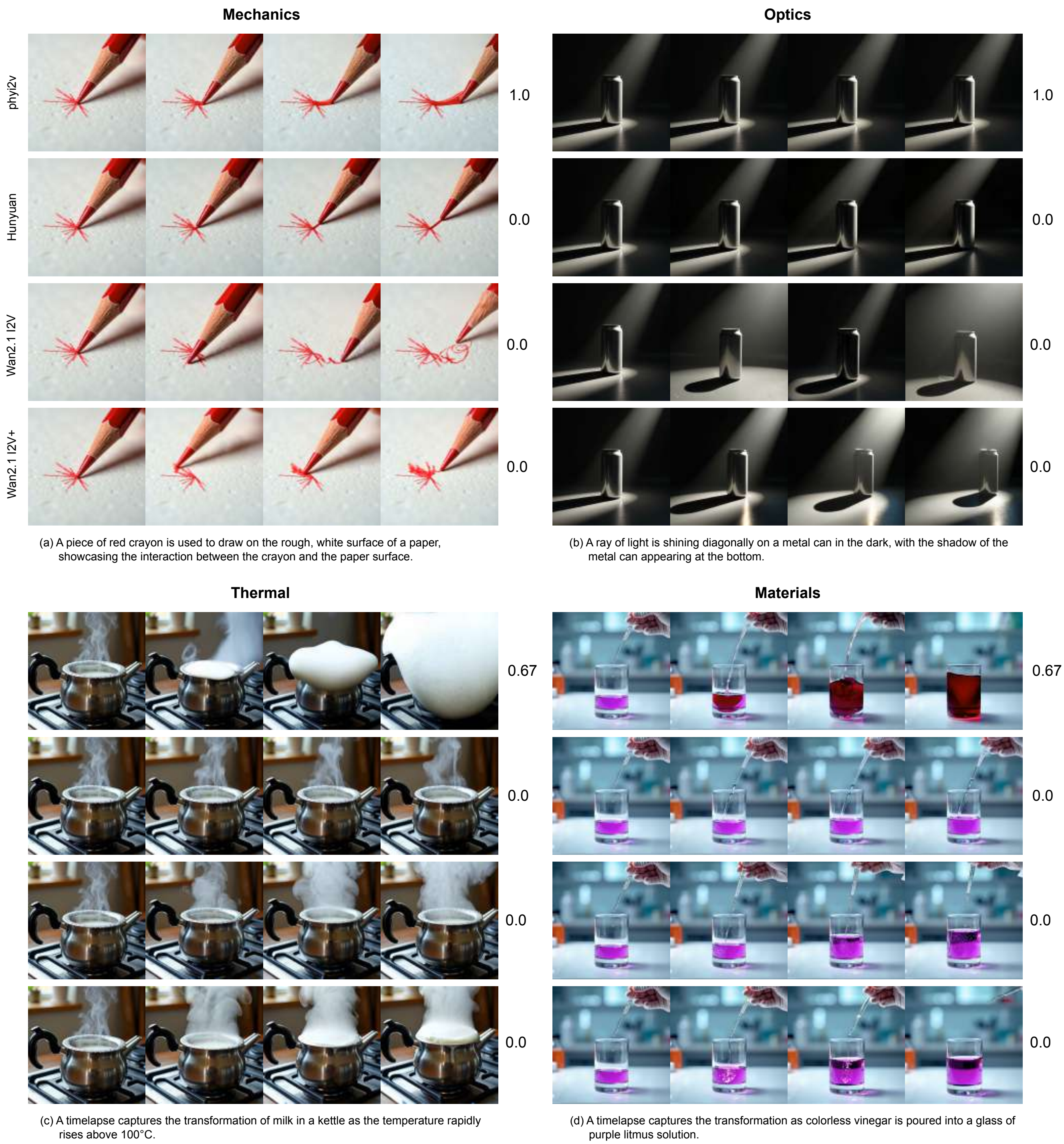
(d) On an airport runway, a leopard knocks over a bottle with its paw, sending it rolling across the asphalt.



(e) Watch as the pink eraser glides across the paper, magically fading the jungle scene and leaving behind a trail of shavings.

(f) Watch as the colorful scarf dances through the air, weaving itself intricately around the lily's petals and stems against the backdrop of a cascading waterfall.

Figure 12: **Qualitative comparison of SGS against naive sampling methods.**



(a) A piece of red crayon is used to draw on the rough, white surface of a paper, showcasing the interaction between the crayon and the paper surface.

(b) A ray of light is shining diagonally on a metal can in the dark, with the shadow of the metal can appearing at the bottom.

(c) A timelapse captures the transformation of milk in a kettle as the temperature rapidly rises above 100°C.

(d) A timelapse captures the transformation as colorless vinegar is poured into a glass of purple litmus solution.

Figure 13: **Qualitative comparison on PhyGenBench.** While our model produces plausible videos in (a-c), only PhyI2V receives high scores, suggesting a scoring bias where its VLM-based refinement games the VLM-based evaluator. In contrast, PhyI2V's high score in (d) stems from a valid advantage in world knowledge (litmus solution chemistry).

F System Prompts

F.1 Taxonomy Guided Prompt Generator

```
Context:
- Object 1: '<Object1_Name>' (Category: <Object1_Taxonomy_Path>)
- Object 2: '<Object2_Name>' (Category: <Object2_Taxonomy_Path>)

Task:
First, evaluate if a meaningful interaction between Object 1 and Object 2 is likely in this scene. The objects were chosen randomly, so they might not make sense together.

If a meaningful interaction exists:
1. Suggest *one* such single interaction verb or short phrase (like 'Cut', 'Pour', 'Collide_with', 'Stack_on') describing how Object 1 might interact with Object 2 in this scene.
2. Describe the primary **major state change** resulting from this interaction, ensuring a significant alteration in the shape or condition of at least one object (e.g., 'Object 2 shatters', 'Object 1 melts onto Object 2', 'Object 2 is torn apart'). Focus on a visually impactful consequence.

Respond ONLY in the strict format: Interaction: [Interaction Name], StateChange: [Description of state change]
Example: Interaction: Dig, StateChange: A hole appears in the ground
If no meaningful interaction seems likely, respond exactly with 'None'.
```

Prompt 1: Object-Centric Tuple Generation.

```
Context:
- Interaction: '<Interaction_Name>' (Category: <Interaction_Taxonomy_Path>)

**IMPORTANT**: The following combinations have been previously generated for this interaction:
- Object1: <Prev_O1_Name>, Object2: <Prev_O2_Name>, StateChange: <Prev_StateChange_Desc>
- ... (more examples if they exist) ...

You MUST suggest a completely different combination. Avoid repeating any of the above Object1, Object2, Scene combinations. Think of alternative scenarios, different object types, or different settings where this interaction could occur.

Task: Suggest a creative and plausible scenario for the given interaction.
1. Suggest an Object 1 (typically the actor or initiator).
2. Suggest a distinct Object 2 (typically acted upon or involved).
3. Describe the primary **state change** resulting from this interaction. This description **must** indicate a clear and definite change in the shape, condition, or relative position of at least one object (e.g., 'Object2 is broken into two pieces', 'A hole appears in Object2'). Focus on the most direct, observable consequence.

Respond ONLY in the strict format: Object1: [Name], Object2: [Name], Scene: [Name], StateChange: [Description of state change]
Example response: Object1: Dog, Object2: Ground, Scene: Backyard, StateChange: A hole appears in the ground
```

Prompt 2: Interaction-Centric Tuple Generation.

```
You are an expert assistant. Your primary goal is to generate two visually rich and detailed image prompts based on the provided scenario details: 'Prompt_Before_Interaction' and 'Prompt_After_Interaction'. The prompts should describe a realistic and well-composed scene.

For both prompts:
- Use descriptive, evocative language. Focus on creating a complete, believable scene.
- Describe subject(s), their state, their relationship to the environment.
- Avoid any conversational filler, questions, or explanations *in your generated prompts*.
```

```
- **IMPORTANT: Avoid negative terms like "not", "no", "without", "absent", "missing", "lack of", etc. Image generation models struggle with negative concepts. Instead, describe what IS present and visible.**

'Prompt_Before_Interaction': Describe a complete, static scene, paying close attention to realistic object placement and a coherent environment.
'Prompt_After_Interaction': Describe the scene *after* the interaction for an image editing model. Clearly depict the final state, ensuring the 'Expected State Change' is visually and dramatically represented.

Adhere strictly to the detailed scenario context, plausibility mode instructions, and JSON output format that will be provided in the subsequent parts of the full instruction set you receive.

---
Scenario Context:
- Object 1 (Actor/Initiator): <Object1_Name> (<Object1_Taxonomy_Path>)
- Object 2 (Acted Upon): <Object2_Name> (<Object2_Taxonomy_Path>)
- Interaction: <Interaction_Name> (<Interaction_Taxonomy_Path>)
- Expected State Change: <StateChange_Description>

- Important: If multiple instances of Object 1 or Object 2 are involved in the interaction, explicitly specify the number of objects in both before and after prompts.

Generate two detailed prompts based on the scenario, focusing on *physical plausibility and realism*:

1.  **Prompt_Before_Interaction:** A detailed prompt for generating a static, photorealistic image. Describe a complete scene with realistic lighting, shadows, and composition. **Crucially, all objects must be realistically placed within the scene (e.g., on a surface, held by a person). Objects must NOT be floating, levitating, or positioned in a physically impossible way.** Describe the background and the spatial relationship between objects to create a believable and interesting context.

2.  **Prompt_After_Interaction:** A detailed instruction prompt for an *image editing* model. Describe the physically plausible result of the interaction. Clearly state the final positions and conditions of the objects, **and ensuring the Expected State Change is visually represented in a compelling way**. Focus on realistic changes to the scene. Use positive descriptive language - describe the new state rather than what is no longer present.

If human involvement is typically required for this interaction, imply or explicitly describe the necessary human action to make the scene realistic.

Output the results in JSON format, exactly like this:
{
  "Prompt_Before_Interaction": "[brief realistic before image prompt]",
  "Prompt_After_Interaction": "[brief realistic after edit instruction incorporating state change]"
}
```

Prompt 3: Physically Plausible Start and End Prompt Generation.

```
Generate two creative and visually rich prompts based on the scenario, suitable for a *cinematic or animated context*:

1.  **Prompt_Before_Interaction:** Describe a visually interesting scene *just before* the interaction. Emphasize dynamic composition. Objects can be anthropomorphized or positioned unusually for storytelling effect, but the scene should still be artistically coherent. Use positive descriptive language.

2.  **Prompt_After_Interaction:** Describe the scene *after* the interaction for an *image editing* prompt. Clearly state the final positions of the objects, showing the result of the interaction in a visually striking or story-driven way, **and ensuring
```

```
    the Expected State Change is represented**. The interaction might be autonomous or
    stylized. Focus on visual storytelling. Use positive descriptive language.

Human involvement is optional; if absent, describe the objects acting with clear intent
    or purpose.
}}
```

Prompt 4: Cinematic and creative Start and End Prompt Generation. Replace L23-30 in Prompt 3.

F.2 Video Prompt Generator

```
Your task is to generate a single, concise video prompt OR filter the sample if the input
     is invalid.

Input Analysis:
1. Critically evaluate the provided 'before' and 'after' images, the scenario details,
    and the image generation prompts.
2. Check for:
     - Consistency: Do the images reasonably match the scenario details (objects,
    interaction, state change) and the image prompts?
     - Logical Transition: Does the change from the 'before' to the 'after' image
    plausibly represent the specified interaction and result in the **Intended State
    Change (<StateChange_Description>)**?
     - Image Quality/Clarity: Are the images clear enough to understand the scene and the
    interaction?

Filtering Condition:
- If you determine that the input is inconsistent, the images are too low quality/unclear
    , the described interaction doesn't match the visual change (especially the specified
     state change (<StateChange_Description>)), or the scenario is nonsensical based on
    the provided images and text, respond *only* with the exact string: `FILTER_SAMPLE`

Video Prompt Generation (if input is valid):
- If the input passes your evaluation, generate a single, concise, and motion centric
    video prompt (single sentence).
- This prompt must describe the dynamic action (<Interaction_Name>) that transforms the '
    before' image into the 'after' image, resulting in the **Intended State Change (<
    StateChange_Description>)**.
- Focus on the action and the resulting state change.
- Maintain consistency with the visual style, objects, and scene depicted in the images.
- Ensure the prompt reflects the requested '<Plausibility_Mode>' plausibility (physical
    realism or cinematic flair).
- **Decision Making:** You must decide whether the interaction requires human
    intervention based on the interaction type and plausibility mode, then adjust your
    prompt accordingly.

**Examples of Human Intervention Decision:**
- Physical Mode: "Bat hits window" -> "A person grabs the baseball bat and swings it
    forcefully, shattering the window glass"
- Cinematic Mode: "Bat hits window" -> "The baseball bat levitates and swings
    autonomously, magically shattering the window"
- Physical Mode: "Ball rolls down hill" -> "A ball rolls down the grassy hill" (no human
    needed)
- Physical Mode: "Knife cuts bread" -> "A person picks up the knife and carefully slices
    through the bread loaf"
- Cinematic Mode: "Knife cuts bread" -> "The knife glides through the air and slices the
    bread by itself"

- Output *only* the generated video prompt as a single string, without any introductory
    text, labels, or explanations.

Scenario Details:
- Object 1: <Object1_Name> (<Object1_Taxonomy_Path>)
- Object 2: <Object2_Name> (<Object2_Taxonomy_Path>)
```

```
- Interaction: <Interaction_Name> (<Interaction_Taxonomy_Path>)
- Intended State Change: **<StateChange_Description>**
- Plausibility Mode: <Plausibility_Mode>

Image Generation Prompts Used:
- Before Image Prompt: <Before_Prompt_Text>
- After Image Prompt: <After_Prompt_Text>

Here are the images:
```

Prompt 5: Video Prompt Generation.

F.3 Video Evaluator

```
Generated Video Evaluation Guidelines (Object Interaction Focused - Score Based & Prompt Considered)
Objective: These guidelines are designed to quantitatively evaluate the quality of object interactions within generated videos. The focus is on assessing how naturally and realistically the video portrays interactions, the smoothness of temporal flow, and how well it reflects the intent of the provided text prompt.

Evaluator: You will watch the generated video, review the accompanying text prompt, and conduct the evaluation based on the criteria and scoring scale below.

Evaluation Criteria and Scoring Scale:

Please watch each video and assign a score from 1 to 5 for each of the following three criteria. When evaluating, consider the content of the text prompt used for video generation to assess the intent and implementation of the interaction.

1. Presence and Clarity of Interaction (1-5 points):
* (Prompt Consideration): If the prompt specified a particular interaction (e.g., "A hits B," "A pushes B," "A and B collide"), is this specific interaction visibly depicted in the video? Is the mechanism of interaction (the contact, the force transfer) clearly represented according to the prompt's intent, rather than just implied by the outcome?
* 1 point: No interaction between the specified objects is depicted, or only negligible, unrelated movements occur. The prompt's interactive intent is entirely absent.
* 2 points: Interaction is attempted or implied as per the prompt, but the critical moment of interaction (e.g., contact, force transfer) is missing, glossed over, or fundamentally flawed (e.g., objects appear to affect each other without clear contact, effects are misaligned with supposed actions, or objects pass through each other when contact is expected). The outcome might occur (e.g., an object falls), but the mechanism described in the prompt is not actually depicted, making the intent poorly reflected.
* 3 points: Interaction is depicted, and the general intent is understandable. However, key aspects of the interaction process (the 'how') are unclear, briefly obscured, or slightly misaligned. For instance, contact might occur, but it's too quick to properly assess, or partially hidden in a way that makes the exact nature of the engagement ambiguous. Prompt intent is partially reflected.
* 4 points: Clear interaction is depicted. It's relatively easy to understand which objects are interacting and how they are physically engaging (e.g., contact is visible and plausible). The mechanism of interaction is mostly clear, and the prompt intent is mostly well reflected.
* 5 points: The complete process of interaction between two or more objects, as specified or implied by the prompt, is very clearly and unambiguously depicted. The mechanism of interaction (e.g., pushing, pulling, colliding, contact points) is visually explicit, sustained enough to be observed, and entirely consistent with the prompt's intent.

2. Physical Plausibility of Video (1-5 points):
* (Prompt Consideration): By default, judge only how well the video obeys everyday physical laws-ignore any prompt or intended effects.
* (Exception): If the prompt explicitly calls for non-standard or "magical" physics, then judge how consistently the video realizes that specified "magic" or special effect.
```

```
* 1 point: The video completely defies physical laws and is highly unnatural (e.g., ignoring gravity, objects passing through each other, unrealistic deformations).
* 2 points: Many physically awkward aspects. The sense of weight, material properties, etc., of the objects is barely noticeable.
* 3 points: Some physically awkward parts exist, but overall it doesn't significantly deviate from common sense. Basic collision, movement, etc., are implemented.
* 4 points: The video appears mostly physically plausible. Object movements, velocity changes, etc., are relatively natural. Minor awkwardness might be present.
* 5 points: The video aligns very well with real-world physics. Gravity, friction, reaction upon collision, object mass/material properties appear naturally reflected.

3. Interaction State Change Causality (1-5 points):
* (Prompt Consideration): Does the video clearly demonstrate that the change in an object's state (e.g., B moving, breaking, changing color, etc.) is a direct and understandable result of the interaction with A, as described or implied by the prompt? Does the prompt specify a particular resulting state change, and is this causal link evident?
* 1 point: The state change of the interacted object appears random, spontaneous, or completely unrelated to the depicted interaction. The prompt's implied or stated consequence of the interaction is not causally linked to the interaction itself.
* 2 points: A state change occurs in the interacted object, but the causal link to the interaction is very weak, highly ambiguous, or seems to be coincidental rather than a direct result. The prompt's intended outcome feels disconnected from the interaction shown.
* 3 points: The interaction leads to a state change in the object, but the causality is not entirely clear or immediate. There might be other distracting elements, or the exact moment/reason for the state change is somewhat obscure, making the link to the prompt's intended consequence partially unclear.
* 4 points: The change in the object's state is clearly and directly caused by the interaction. The "before and after" states are distinct, and the interaction serves as a convincing trigger, mostly aligning with the prompt's implied or explicit causal chain.
* 5 points: The video perfectly illustrates the cause-and-effect relationship. The interaction unequivocally and visibly leads to the specific change in the object's state as described or logically implied by the prompt. The sequence of interaction leading to the resultant state change is unambiguous and compelling.

4. Temporal Continuity and Absence of Unnatural Jumps (1-5 points):
* (Prompt Consideration): Unless the prompt intentionally requested scene transitions or effects, is the temporal flow natural within the depicted segments?
* 1 point: Severe issues throughout the video, such as frame drops, scene jumps that disrupt the depicted action, objects teleporting or drastically changing shape unnaturally.
* 2 points: Unnatural jumps between frames or abnormal object movements (e.g., flickering, warping unrelated to interaction) are frequently noticeable, disrupting the viewing flow of the depicted scenes.
* 3 points: Minor jumps or unnatural movements appear intermittently, but they don't significantly hinder understanding the flow of the action that is shown.
* 4 points: The temporal flow of the depicted action is generally smooth with almost no jumps. Object movements are continuous.
* 5 points: Video playback is very smooth, and the temporal flow from the beginning to the end of the depicted interaction is completely natural. No frame drops or abnormal object movements are observed.

Evaluation Method:

First, review the text prompt provided with each video.

Watch the video and assign a score between 1 and 5 for each of the three criteria (1. Presence/Clarity of Interaction, 2. Physical Plausibility, 3. Interaction State Change Causality, 4. Temporal Continuity). Evaluate by considering the prompt content.

Briefly noting the reason for each score or referencing specific video segments (timestamps) and their relevance to the prompt content helps improve evaluation
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    reliability. (e.g., "Interaction Clarity 3 points - Prompt requested 'ball knocking
    over a cup', but the ball just passes by the cup (0:07s).")

Reporting Results:

Record the scores for the four criteria for each video in the provided format (e.g.,
    spreadsheet, evaluation document).

Sum the scores for each criterion to calculate the Total Score (minimum 4 points ~
    maximum 20 points) and record it as well.

If necessary, you can add brief comments on the video's implementation level relative to
    the prompt and the overall quality of the interaction alongside the total score.

We hope these guidelines facilitate consistent and quantitative video evaluations. Thank
    you for your participation in the evaluation.

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Please read this instruction. After this, I will provide the video and its prompt.
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Prompt 6: Gemini video evaluation prompt.

G Future Work

Our research will focus on two key areas: leveraging a more robust understanding of physics in the model and broadening the framework's practical applications. To expand the model's current implicit knowledge, we plan to directly integrate simplified physics engines. These engines will guide the video generation process, ensuring that physical laws are respected from the beginning rather than merely filtering out implausible results. Besides, we plan to expand our taxonomy beyond its current 1,300 objects and 500 interactions. Generating a much larger dataset than our final training set of 1,525 videos will allow us to employ more advanced preference tuning techniques, building on our initial use of human-labeled data and enabling us to better capture the nuances of realistic interactions.

For this work to have a broader impact, it must be both accessible and applicable. Therefore, we will utilize the open-source image editors and VLMs. This will address the current limitation of relying on a single third-party model. An open framework can accelerate progress in critical areas such as robotics, where a core goal is to equip world models with a robust understanding of physical cause and effect. In addition to robotics, these advancements will enable the creation of dynamic and interactive content for virtual reality and other immersive 4D applications.